Team Members

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Project Description

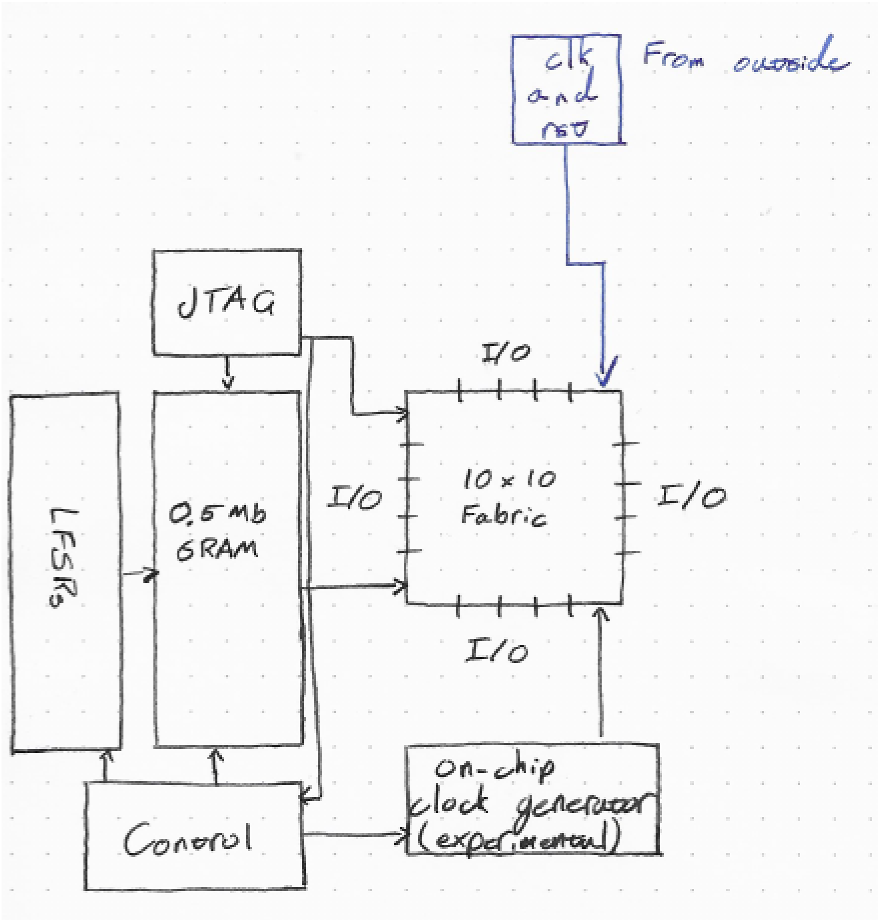
From-scratch basic FPGA, consisting of CLBs, configurable short-distance routing, configurable long-distance routing, and configurable I/O.

- **External Interface**
    - JTAG interface
    - GPIO pins (the number of power and ground pins required will determine how many GPIOs, but we anticipate around 16)
    - All fabric state elements will be configured into a single scan-chain
    - Separate JTAG instructions for fabric scan-chain access, on-chip buffer access, and control unit access
- **Bitstream Creation**
    - Synthesis and Mapping to the custom FPGA fabric will be generated with the open-source VTR toolchain. Conversion from VTR output to loadable bitstream wil be either implemented or sourced.
- **Microarchitecture**
    - CLB: two 3LUTs in a fracturable configuration, a flip-flop, and a carry chain to optimize adders
    - Configurable I/O: will allow selection of input, output, or inout
    - Routing: Using switch and connection boxes with skip routing for long distance communication
- **Input Data**
    - The FPGA will run offline, with data loaded into an on-chip buffer
    - We will include LFSRs to allow testing power with random input data
- **Clocking**
    - Input clock at 250 MHz
    - JTAG clock at 10 MHz
    - Internal clock generator as an experiment
- **Extras if we finish *very* early**
    - Hard Macros (e.g. MAC DSP Units)
    - Fancy Clocking (PLL, DLL, &c.)

Target Specs:

Area: 1 mm^2
CLBs: 100 (10x10)
GPIOs: ~16
Frequency: 250 MHz
Power: 100 mW
On-chip memory: 0.5 Mb

Project Block Diagram and Partitioning



Subsystem specs:

- Tile area (CLB + routing per CLB):
  - 4000 um^2
- Fabric area (total):
  - 0.5 mm^2
- SRAM area:
  - 0.25 mm^2
- Leftover area:
  - 0.25 mm^2 for LFSRs, clock generators, JTAG, routing, and margin

Partitioning:

- Fabric is datapath
- Control/JTAG/LFSRs are control
- SRAM is array

Background References

Jin Hee Kim and Jason H. Anderson. 2017. Synthesizable standard cell FPGA fabrics targetable by the Verilog-to-Routing CAD flow. ACM Trans. Reconfigurable Technol. Syst. 10, 2, Article 11 (April 2017), 23 pages.

Jason Luu, Jeffrey Goeders, Michael Wainberg, Andrew Somerville, Thien Yu, Konstantin Nasartschuk, Miad Nasr, Sen Wang, Tim Liu, Nooruddin Ahmed, Kenneth B. Kent, Jason Anderson, Jonathan Rose, and Vaughn Betz. 2014. VTR 7.0: Next generation architecture and CAD system for FPGAs. ACM Trans. Reconfig. Technol. Syst. 7, 2, Article 6 (June 2014), 30 pages.

Aken'Ova, Victor, and Resve Saleh. "A "soft++" eFPGA physical design approach with case studies in 180nm and 90nm." *IEEE Computer Society Annual Symposium on Emerging VLSI Technologies and Architectures (ISVLSI'06)*. IEEE, 2006.

Project Schedule: